

Ronald A. Ruffin

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PROFESSIONAL PROFILE

Senior Software Development Engineer with 20+ years of experience supporting, modernizing, and enhancing enterprise Java and .NET applications in secure, regulated environments. Extensive background in backend business logic, database systems, security modules, and automated testing within n-tier architectures.

Experienced in leading modernization initiatives, improving version control practices, implementing regression automation, and reducing technical debt in legacy systems.

Over the past several years, have integrated AI-assisted development workflows into both enterprise and personal projects — using large language models and GitHub Copilot to accelerate prototyping, refine architecture, strengthen unit testing, and compress feedback cycles while maintaining full engineering ownership and validation. My AI-augmented process is not “ask a chatbot for answers.” It is a disciplined loop: isolate the evidence, ask targeted questions, inspect the reasoning, test the result, and only then share the fix.

Known for disciplined testing, architectural clarity, and thoughtful system evolution rather than rapid but fragile delivery.

Built and deployed a publicly inspectable symbolic math application to demonstrate architectural rigor and AI-augmented SDLC practices beyond proprietary enterprise codebases.

TECHNICAL EXPERTISE

Languages:

Java (JSE/JEE), SQL/PL-SQL, JavaScript, Perl, C#, VB.NET, C/C++, JSP, Unix Shell

Frameworks & Platforms:

Spring Boot, Vaadin, PrimeFaces, JSF, JPA/Hibernate, Apache Shiro, Maven, Ant, n-tier architectures

Databases:

Oracle 19c, SQL Server, Sybase, Informix, Pervasive, DB2, Microsoft Access

Testing & Quality:

JUnit, Selenium IDE, Selenium WebDriver, SonarQube, SonarCloud, Checkmarx, Dependency Track (SBOM)

DevOps & Tooling:

GitHub, Bitbucket, Jenkins, Docker, NetBeans, Eclipse, Visual Studio Code, IntelliJ

AI-Augmented Development:

GitHub Copilot, ChatGPT, LLM-assisted architectural prototyping, AI-assisted unit test generation and edge-case analysis

PROFESSIONAL EXPERIENCE

Empower AI, Inc.

(Formerly NCI Information Systems, Inc.)

U.S. Air Force AFLCMC/SSDA Application Team

Wright Patterson AFB

Dayton, OH

Web Software Developer V | September 2020 – Present

Support and enhance the SSDA Web Security Module, a web-based security access framework used by a federal client. Perform maintenance, bug fixes, and feature enhancements across front-end, middleware, and back-end layers using Java, Spring Boot/Vaadin, PrimeFaces, JSF, JPA/Hibernate, Apache Shiro, Oracle 19c, and JBoss.

- Lead migration and modernization initiatives, including introduction of Git-based version control and repository management for legacy codebases.
- Design and implemented automated regression testing using Selenium IDE and Selenium WebDriver, integrating with Zephyr and CI workflows.
- Prototype modernization efforts to address SBOM findings, including migration from Ant to Maven and exploration of Spring Boot/Vaadin architecture.
- Develop report-generation subsystem using JasperSoft Studio.
- Integrate context-based help system using DrExplain.

AI-Augmented Development Practices:

- Incorporate generative AI tools (ChatGPT, GitHub Copilot) to accelerate prototype development and explore modernization paths.
- Use AI-assisted code scaffolding and refactoring to evaluate alternative architectural approaches during framework migration efforts.
- Leverage AI to generate test cases, identify edge conditions, and validate refactor impacts during legacy-to-modern transitions.
- Maintain full architectural ownership and manual validation in secure, compliance-driven environments.

PUBLIC & INDEPENDENT ENGINEERING PROJECTS

The Algebrator

Personal Project | March 2025 – Present Public GitHub Repository | Deployed Application

Designed and deployed a symbolic algebra and quadratic problem-solving application to demonstrate architectural clarity, disciplined testing practices, and AI-augmented software development workflows.

- Architected modular Java-based solution integrating symbolic math engine (Symja) with custom business logic layers.
- Designed intuitive calculator-style UI with function panels, variable handling, and mode toggles (e.g., degree/radian support).
- Implemented extensive built-in unit testing to validate arithmetic, trigonometric, and symbolic operations, including round-trip verification scenarios.
- Iteratively refined UI/UX and system design based on controlled experimentation and specification-driven validation.
- Deployed application to cloud hosting environment for public demonstration.

AI-Augmented Workflow:

- Used AI as a structured development collaborator to brainstorm architectural options, prototype alternative implementations, and generate scaffolding code.
- Directed iterative refinement cycles by reviewing AI-generated solutions, enforcing design constraints, and validating correctness through manual testing and unit coverage.
- Treated AI as a high-speed junior engineer — setting objectives, reviewing output, and maintaining full architectural ownership.

Custom MIDI Pedal Controller (Arduino-Based)

Designed and implemented a custom embedded MIDI controller to translate physical pedal inputs into virtual instrument control signals.

- Designed and wired custom circuitry.
- Programmed Arduino microcontroller in C++.
- Implemented MIDI protocol handling and signal mapping.
- Debugged hardware/software interaction issues.
- Recently enhanced design using AI-assisted iteration to refine signal handling and firmware logic.

PROFESSIONAL EXPERIENCE (CONTINUED)

Octo Consulting Group

U.S. Air Force AFFSO/ELF Application Team

Wright Patterson AFB

Dayton, OH

Senior Application Developer | 2019–2020

Supported ELF financial applications for the U.S. Air Force (Java, SQL Server, myBatis, Wicket). Performed maintenance, enhancements, and middleware modifications. Participated in Agile ceremonies and version-controlled workflows (Subversion, JIRA, Ant).

The Eliassen Group

General Dynamics Information Technology (GDIT)

Dayton, OH

Software Engineer / Test Automation Engineer | 2019

Designed and implemented automated test sequences for Android Tactical Assault Kit (ATAK), using TestComplete (vbScript). Developed Java-based simulation plug-ins in Android Studio. Integrated Jenkins-driven regression automation within Agile sprint cycles.

Oxford Global Resources/ Triple E Partners, Ltd./ Cohesion Corp.

Dayton/Cincinnati area

Software Engineer (Contract Roles) | 2017–2019

Delivered Java/JEE maintenance and enhancement work across automotive, financial, and retail domains.

Key contributions included:

- Middleware and web-service enhancements (Java, SQL, DB2, Oracle)
- POS system transition support and reporting subsystem updates
- Ant-to-Maven build modernization efforts
- Daily collaboration within Agile teams
- Regression and unit test validation prior to QA handoff

Technologies: Java, Spring Boot, WebLogic, IntelliJ, Maven, SQL Server, Oracle, DB2, GitHub.

Jenzabar, Inc.

Cincinnati, OH

Software Engineer (Direct Hire) | 2012–2016

Maintained and enhanced CX Student, a student and faculty-facing portion of JICS, a commercial higher-education portal application.

- Developed Java middleware plugins and server-side processes (Java, Perl, C/C++)
- Enhanced student-facing applications using JavaScript/Dojo
- Maintained Informix stored procedures and database integrations
- Coordinated with in-house and offshore teams

Technologies: Java, JBoss, Unix/Linux, Informix, n-tier architecture.

EARLIER CAREER EXPERIENCE (1990–2011)

Held progressively responsible software engineering and systems roles across federal, commercial, and industrial environments. Contributed to development, modernization, and support of client-server and early web-based enterprise systems.

Highlights:

- Designed and maintained J2EE applications (WebSphere, Struts, Hibernate, Spring, JSP) for federal government systems (2005–2011).
- Implemented PL/SQL stored procedures and enterprise reporting subsystems (JasperReports, iReports).
- Led object-oriented modernization initiatives using UML and Rational Unified Process (RUP).
- Developed and maintained Swing-based client-server applications incorporating JMS, JNDI, and MQ messaging.
- Designed multi-layered APIs in VB6/VB.NET using .NET Interop for industrial automation systems.
- Built object-oriented business rule frameworks in VB6 and C++ within 3-tier architectures.
- Early career included systems engineering, LAN administration, desktop application development, and hardware integration roles.

Technologies across this period included:

Java (J2EE), Spring, Hibernate, Struts, WebSphere, Swing, PL/SQL, Oracle 8–10g, DB2, Informix, VB6, VB.NET, C++, COM/.NET Interop, JMS, XML/XSL, UML, RUP, Solaris, Windows NT/XP.

EDUCATION

Strayer University, Newington, VA

Coursework toward Computer Information Systems (2000–2002)

Control Data Institute, Arlington, VA

Certificate in Computer Systems Maintenance (1985–1986)